HRIDEYA SHARMA

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PROFILE

Computer Engineering undergraduate at Thapar Institute of Engineering and Technology, with interests and skills in Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, and LLMs. Experience in designing and building AI applications like recommendation engines, speech engines, NLP pipelines, and automated data extraction systems. Experience in designing and implementing end-to-end ML solutions using deep learning frameworks like PyTorch and Tensor Flow, transformers, FastAPI, with skills in designing and maintaining powerful back-end systems, REST APIs, and data processing pipelines. Enthusiasm for practical AI research and building smarter systems that solve real-world problems.

EDUCATION

Thapar Institute of Engineering and Technology Patiala, Punjab
B.Tech in Computer Engineering *2024–2028*
CGPA: **9.29/10**; Merit Scholarship Recipient

Major R.N. Kapoor DAV Public Sr. Sec. School *CBSE Class XII: 87% | 2022–2024*

The S.D. Vidya School *CBSE Class X: 93.8% | 2012–2022*

PUBLICATIONS

Less is More in Kolmogorov-Arnold Networks: The Shared-Atom Architecture *CoLoRAI 2026 Workshop, ICML 2026*

PyTorch, Deep Learning, Neural Networks, Scientific Machine Learning

- Proposed Shared-Atom KAN (SA-KAN), a parameter-efficient variant of Kolmogorov-Arnold Networks that learns a shared dictionary of reusable basis functions.
- Introduced sparse atom mixing and low-rank functional factorization to reduce model complexity while maintaining expressiveness.
- Achieved around 30 percent parameter reduction compared to Vanilla KAN while improving performance on structured learning tasks.

RESEARCH EXPERIENCE

PLBench: Benchmarking Logical Reasoning in Large Language Models *Ongoing*

Python, PyTorch, Hugging Face Transformers, LLM Evaluation

- Developed PLBench, a benchmark for evaluating propositional logic reasoning in large language models.
- Generated multi-depth reasoning tasks with formally verifiable proof chains and automated validation pipelines.
- Investigating reasoning failures, hallucinations, and generalization across model scales and prompting methods.

Mechanism-Aware World Models for Robust Representation Learning *Ongoing*

PyTorch, RL, Representation Learning, DMControl

- Investigating mechanism-aware representations for improving world-model generalization under changing environment dynamics.
- Conducting experiments on DMControl environments to evaluate transfer, mechanism extrapolation, and long-horizon prediction.
- Studying the relationship between latent structure learning and robust reinforcement learning performance.

PROJECTS

AI-Powered Meeting Summarizer GitHub

- Developed a platform that converts uploaded meeting audio into structured summaries and actionable insights.
- Implemented Whisper-based transcription and LLM workflows for summaries, decisions, and action-item extraction.
- Built a FastAPI backend and React frontend; applied prompt engineering to improve output quality and consistency. *Technologies: Python, PyTorch, Whisper, FastAPI, NLP, LLMs.*

Movie Recommendation System Using NLP GitHub

- Developed a content-based recommendation engine using TF-IDF vectorization and cosine similarity.
- Combined genres, cast, keywords, and numerical features to retrieve personalized top-N recommendations. *Technologies: Python, Scikit-learn, NLP, TF-IDF.*

Automated E-Commerce Product Data Scraper GitHub

- Built an automated pipeline using Selenium and BeautifulSoup to extract structured product data from dynamic websites.
- Added support for JavaScript-rendered pages, error handling, validation, and structured storage. *Technologies: Python, Selenium, BeautifulSoup.*

TECHNICAL SKILLS

Programming Languages	Python, Java, C/C++
Machine Learning & AI	Machine Learning, Deep Learning, Reinforcement Learning, NLP, LLMs, Recommendation Systems, Generative AI
Frameworks & Libraries	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV, LangChain
Backend & Databases	FastAPI, PostgreSQL
Tools & Technologies	Git, GitHub, REST APIs, Data Processing, Web Scraping

ACHIEVEMENTS

- Code for Good 2026 Participant: Selected for the JPMorgan Chase Code for Good Hackathon 2026, collaborating in a team environment to develop technology-driven solutions for nonprofit organizations.
- Merit Scholarship Recipient: Awarded merit-based scholarship at Thapar Institute of Engineering and Technology for outstanding academic performance.

INTERESTS

Artificial Intelligence ▪ Machine Learning Systems ▪ Reinforcement Learning ▪ Large Language Models ▪ Applied AI Research ▪ Data Engineering